SINTEZA LC DVOPOLA
po Cauueru

I. Cauerov oblik (kanonski)

- stupanj nazivnika > stupanj brojnika → pol u ∞

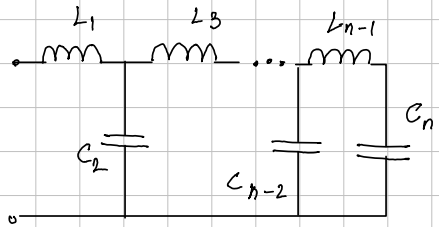
$$W(s) = k_1 \cdot s + \cfrac{1}{k_2 \cdot s + \cfrac{1}{k_3 \cdot s + \cfrac{1}{k_3 \cdot s + \dots \cfrac{1}{k_n \cdot s}}}}$$

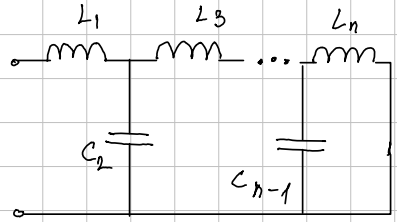
razvoj u niži razlomak

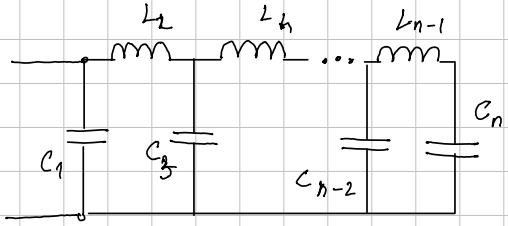
- stupanj nazivnika < stupanj brojnika → nula u ∞

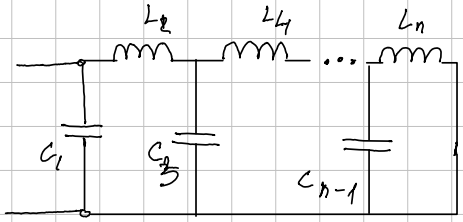
$$W(s) = \cfrac{1}{k_1 \cdot s + \cfrac{1}{k_2 \cdot s + \dots + \cfrac{1}{k_n s}}}$$

→ bez obzira je li W(s) admitencija i impedancija, dvopol ima jedan od ovih oblika









I. Cauerov kanonski oblik LC dvopola: oblik ovisi o polu ili nuli u beskonačnosti ili u nuli

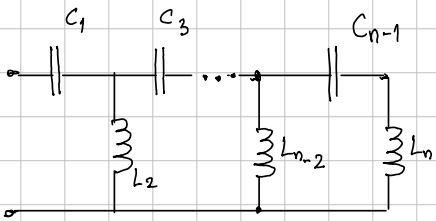
II. Cauerov oblik (kanonski)

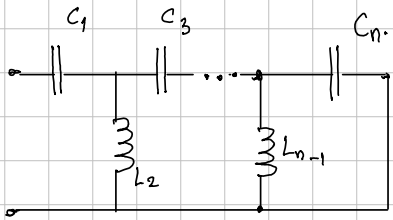
→ polinom brojnika paran
polinom nazivnika neparan
(ima pol u nuli)

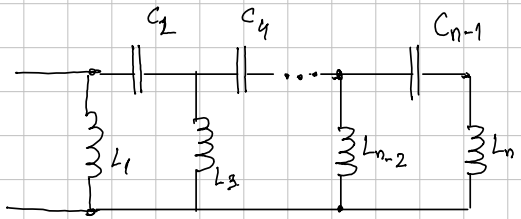
$$W(s) = \frac{k_1}{s} + \cfrac{1}{\frac{k_2}{s} + \cfrac{1}{\frac{k_3}{s} + \cdots + \cfrac{1}{\frac{k_n}{s}}}}$$

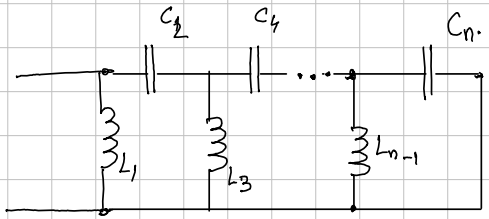
→ brojnik neparan
nazivnik paran

$$W(s) = \cfrac{1}{\frac{k_1}{s} + \cfrac{1}{\frac{k_2}{s} + \cdots + \cfrac{1}{\frac{k_n}{s}}}}$$









Primjeri: Cauer

Primjer 1.) Za zadanu fju impedancije LC dvopola odrediti prvi i drugi kanonski oblik po Cauer-u

PRVI KAN. OBLIK: od najviše do najniže

$$Z_{LC}(s) = \frac{s^3+2s}{s^4+3s^2+1} \begin{matrix} \to \text{neparan} \\ \to \text{paran} \end{matrix} \quad Z_{LC}(s) = \cfrac{1}{\frac{k_1}{s} + \cfrac{1}{\frac{k_2}{s} + \cfrac{1}{\frac{k_3}{s} + \cfrac{1}{\frac{k_4}{s}}}}}$$

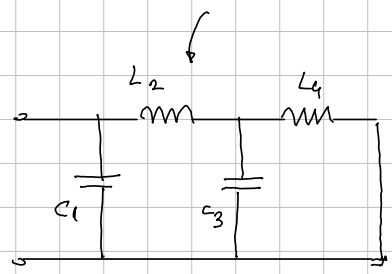
dobro dobiti izraziti : $Z_{LC}(s) = \frac{1}{Y_u(s)} = \frac{1}{\frac{s^4+3s^2+1}{s^3+2s}}$

$$(s^4+3s^2+1) : (s^3+2s) = s \to C_1 = 1$$
$$\frac{-s^4 - 2s^2}{0}$$
$$+s^2 + 1$$

$$Z_{LC}(s) = \cfrac{1}{s + \frac{s^2+1}{s^3+2s}} = \cfrac{1}{s + \cfrac{1}{\frac{s^3+2s}{s^2+1}}} \Rightarrow (s^3+2s):(s^2+1) = s \to L_2 = 1$$
$$\frac{-s^3 - s}{s}$$

$$Z_{LC}(s) = \cfrac{1}{s + \cfrac{1}{s + \frac{s}{s^2+1}}} = \cfrac{1}{s + \cfrac{1}{s + \cfrac{1}{\frac{s^2+1}{s}}}} \Rightarrow (s^2+1):s = s \to C_3 = 1$$
$$1$$

$$Z_{LC}(s) = \cfrac{1}{s + \cfrac{1}{s + \cfrac{1}{s + \frac{1}{s}}}} \to L_4 = 1$$



$C_1 = 1, \quad L_2 = 1, \quad C_3 = 1, \quad L_4 = 1$

DRUGI KAN OBLIK od najniže do najviše

$$Z_{LC}(s) = \frac{s^3+2s}{s^4+3s^2+1} = \frac{2s+s^3}{1+3s^2+s^4} \longrightarrow Z_{LC}(s) = \frac{1}{\frac{1+3s^2+s^4}{2s+s^3}}$$

$$(1+3s^2+s^4) : (2s+s^3) = \frac{1}{2s} \to \boxed{L_1 = 2}$$
$$\frac{-1 - \frac{1}{2}s^2}{\frac{5}{2}s^2 + s^4}$$

$$\longrightarrow Z_{LC}(s) = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{\frac{5}{2}s^2+s^4}{2s+s^3}}} = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{2s+s^3}{\frac{5}{2}s^2+s^4}}}$$

$$(2s+s^3) : \left(\frac{5}{2}s^2+s^4\right) = \frac{4}{5s} \to \boxed{C_2 = \frac{5}{4}}$$
$$\frac{-2s - \frac{4}{5}s^3}{\frac{1}{5}s^3}$$

$$\longrightarrow Z_{LC}(s) = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{4}{5s} + \frac{\frac{1}{5}s^3}{\frac{5}{2}s^2+s^4}}} = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{4}{5s} + \cfrac{1}{\frac{\frac{5}{2}s^2+s^4}{\frac{1}{5}s^3}}}}$$

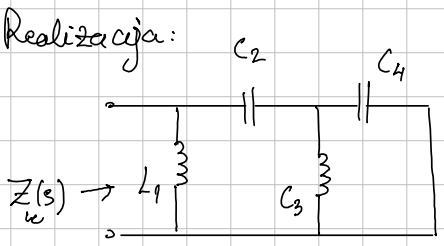
$$\left(\frac{5}{2}s^2 + s^4\right) : \left(\frac{1}{5}s^3\right) = \frac{25}{2s} \to \boxed{L_3 = \frac{2}{25}}$$
$$s^4$$

$$\Longrightarrow Z_{LC}(s) = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{4}{5s} + \cfrac{1}{\frac{25}{2s} + \frac{s^4}{\frac{s^3}{5}}}}} = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{4}{5s} + \cfrac{1}{\frac{25}{s} + \cfrac{1}{\frac{s^3}{\frac{5}{s^4}}}}}}$$

$$\frac{s^3}{5} : \frac{1}{s^4} = \frac{1}{5s} \to \boxed{C_4 = 5}$$

$$Z_{LC}(s) = \cfrac{1}{\frac{1}{2s} + \cfrac{1}{\frac{4}{2s} + \cfrac{1}{\frac{25}{2s} + \cfrac{1}{\frac{1}{5s}}}}} \Longrightarrow \cfrac{1}{\frac{1}{L_1 s} + \cfrac{1}{\frac{1}{C_2 s} + \cfrac{1}{\frac{1}{L_3 s} + \cfrac{1}{\frac{1}{C_4 s}}}}}$$

Realizacija:



$$L_1 = 2 \quad C_2 = \frac{5}{4} \quad L_3 = \frac{2}{15} \quad C_4 = 5$$

SINTEZA RC PO CAUERU

I. Cauerov kan. oblik

$\boxed{Z}$ st. brojnika = st. nazivnika

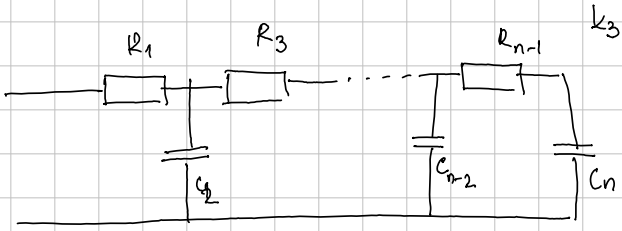
$$Z_{RC}(s) = k_1 + \cfrac{1}{k_2 s + \cfrac{1}{k_3 + \dots + \cfrac{1}{k_n (s)^*}}}$$

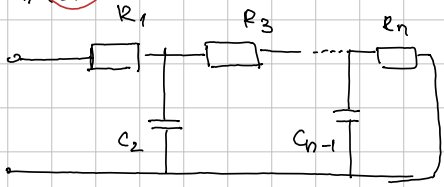
2. slučaj

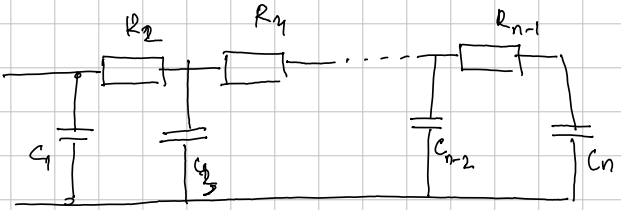
$\boxed{Y}$ st. br < st. naz.

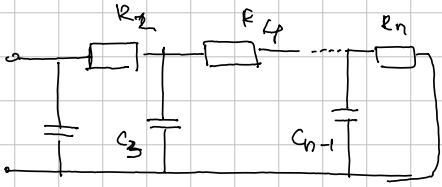
$$Y_{RC}(s) = k_1 s \cfrac{1}{k_2 + \cfrac{1}{k_3 s + \dots + \cfrac{1}{k_n (s)^*}}}$$

2. slučaj









II. Cauerov kan. oblik

$\boxed{Z}$ st. najmanji potenc. br < st. ... naz.

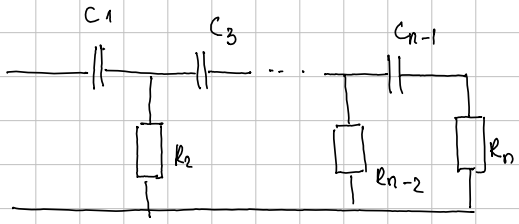
$$Z_{RC}(s) = \frac{k_1}{s} + \cfrac{1}{k_2 + \cfrac{1}{\frac{k_3}{s} + \dots + \cfrac{1}{k_n (s)^*}}}$$

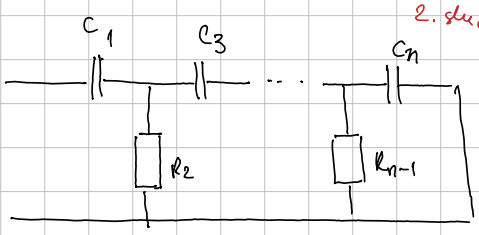
2 slučaj

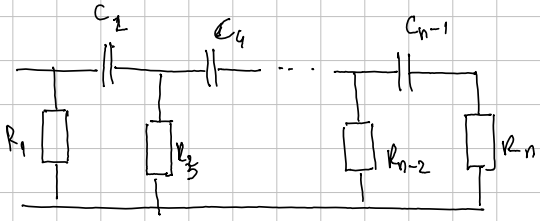
$\boxed{Y}$ st. uz najmanji potenc. br. i naz =

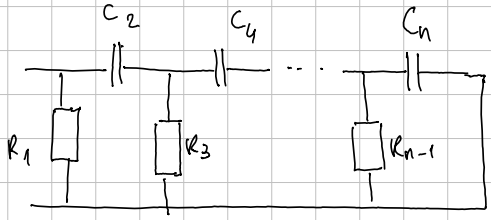
$$Y_{RC}(s) = k_1 + \cfrac{1}{\frac{k_2}{s} + \cfrac{1}{k_3 + \dots + \cfrac{1}{k_n (s)^*}}}$$

2. slučaj









Primjeri Cauer

Primjer 1) Realizirati prvom Cauerovom realizacijom dvopol čija je ulazna imp.

$$Z(s) = \frac{2s^2+14s+20}{s^2+4s+3}$$

st. naz = st. br.

$$(2s^2+14s+20):(s^2+4s+3) = 2$$
$$\underline{-2s^2-8s-6}$$
$$0 \quad +6s+14$$

$$\boxed{R_1=2}$$

$$Z_{RC}(s) = 2+\frac{6s+14}{s^2+4s+3} = 2+\frac{1}{\frac{s^2+4s+3}{6s+14}}$$

$$(s^2+4s+3):(6s+14) = \frac{1}{6}s$$
$$\underline{-s^2-\frac{7}{3}s}$$
$$\frac{5}{3}s+3$$

$$Z_{RC}(s) = 2+\frac{1}{\frac{1}{6}s+\frac{\frac{5}{3}s+3}{6s+14}} = 2+\frac{1}{\frac{1}{6}s+\frac{1}{\frac{6s+14}{\frac{5}{3}s+3}}}$$

$$\boxed{C_2=\frac{1}{6}}$$

$$(6s+14):\left(\frac{5}{3}s+3\right) = \frac{18}{5}$$
$$\underline{-6s-\frac{27}{5}}$$
$$\frac{16}{5}$$

$$Z_{RC}(s) = 2+\frac{1}{\frac{1}{6}s+\frac{1}{\frac{16}{5}}} \quad 2+\frac{1}{\frac{1}{6}s+\frac{1}{\frac{18}{5}+\frac{1}{\frac{\frac{5}{3}s+3}{\frac{16}{5}}}}}$$

$$\frac{18}{5}+\frac{\frac{5}{3}s+3}{\frac{16}{5}}$$

$$\boxed{R_3=\frac{18}{5}}$$

$$\left(\frac{5}{3}s+3\right):\left(\frac{16}{5}\right) = \frac{25}{48}s$$
$$\cancel{-\frac{5}{3}s}+3$$

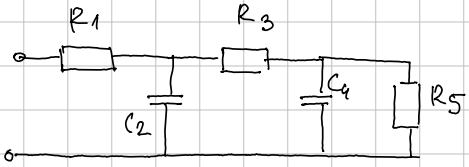
$$Z_{RC} = 2+\frac{1}{\frac{1}{6}s+\frac{1}{\frac{18}{5}+\frac{1}{\frac{25}{48}s+\frac{3}{\frac{16}{5}}}}} = 2+\frac{1}{\frac{1}{6}s+\frac{1}{\frac{18}{5}+\frac{1}{\frac{25}{48}s+\frac{1}{\frac{16}{15}}}}}$$

$$\boxed{C_4=\frac{25}{48}}$$

$$\boxed{R_5=\frac{16}{15}}$$

Realizacija:

$$Z_{RC}(s) = R_1+\frac{1}{sC_2+\frac{1}{R_3+\frac{1}{sC_4+\frac{1}{R_5}}}}$$



$R_1=2$ $\quad R_3=\frac{18}{5}$ $\quad R_5=\frac{16}{15}$

$C_2=\frac{1}{6}$ $\quad C_4=\frac{25}{48}$

Primjer 3.) Druga Cauerova realizacija

$$Z(s) = \frac{2s^2+14s+20}{s^2+4s+3}$$, jednaki stupnjevi → Y

$$Y_{RC}(s) = \frac{1}{\frac{s^2+4s+3}{2s^2+14s+20}} = \frac{1}{\frac{3+4s+s^2}{20+14s+2s^2}}$$

$$(3+4s+s^2):(20+14s+2s^2) = \frac{3}{20}$$

$$\frac{19}{10}s + \frac{7}{10}s^2$$

$$Y_{RC}(s) = \frac{1}{\frac{3}{20} + \frac{\frac{19}{10}s + \frac{7}{10}s^2}{20+14s+2s^2}} = \frac{1}{\frac{3}{20} + \frac{1}{\frac{20+14s+2s^2}{\frac{19}{10}s+\frac{7}{10}s^2}}}$$

$$\boxed{R_1 = \frac{3}{20}}$$

$$(20+14s+2s^2):\left(\frac{19}{10}s+\frac{7}{10}s^2\right)$$

$$\frac{126}{19}s+2s^2 \qquad = \frac{200}{19s}$$

$$Y_{RC}(s) = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{\frac{126}{19}s+2s^2}{\frac{19}{10}s+\frac{7}{10}s^2}}} = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{1}{\frac{\frac{19}{10}s+\frac{7}{10}s^2}{\frac{126}{19}s+2s^2}}}}$$

$$\boxed{C_2 = \frac{19}{200}}$$

$$\left(\frac{19}{10}s+\frac{7}{10}s^2\right):\left(\frac{126}{19}s+2s^2\right) = \frac{8}{63}s^2 \qquad = \frac{361}{1260}$$

$$Y_{RC}(s) = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{1}{\frac{361}{1260} + \frac{\frac{8}{63}s^2}{\frac{126}{19}s+2s^2}}}} = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{1}{\frac{361}{1260} + \frac{1}{\frac{\frac{126}{19}s+2s^2}{\frac{8}{63}s^2}}}}}$$

$$\boxed{R_3 = \frac{361}{1260}}$$

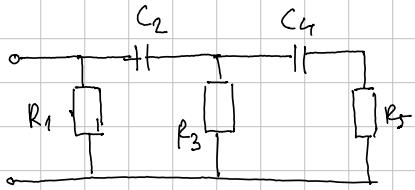
$$\left(\frac{126}{19}s+2s^2\right):\left(\frac{8}{63}s^2\right) = \frac{3969}{76s}$$

$$2s^2$$

$$Y(s) = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{1}{\frac{361}{1260} + \frac{1}{\frac{3969}{76s} + \frac{2s^2}{\frac{8}{63}s^2}}}}} = \frac{1}{\frac{3}{20} + \frac{1}{\frac{200}{19s} + \frac{1}{\frac{361}{1260} + \frac{1}{\frac{3969}{76s} + \frac{1}{\frac{63}{4}}}}}}$$

$$\boxed{C_4 = \frac{76}{3969}}$$

$$\boxed{R_5 = \frac{63}{4}}$$

Realizacija:

$$Y_{RC} = \frac{1}{R_1 + \frac{1}{\frac{1}{C_2 s} + \frac{1}{R_3 + \frac{1}{\frac{1}{sC_4} + \frac{1}{R_5}}}}}$$